

Bo Rui

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Ph.D. Student, Department of Mechanical Engineering, The Pennsylvania State University

Research Interests:

- Mechanical safety of next-generation batteries
- Electro-chemo-mechanical coupling
- Multiphysics modeling (COMSOL, ABAQUS, MATLAB)
- Experimental mechanics of energy storage systems

Professional Memberships:

- Electrochemical Society (ECS) – Student Member
- American Society of Mechanical Engineers (ASME) – Student Member
- SAE International – Student Member

Education:

The Pennsylvania State University– Ph.D. in Mechanical Engineering (Aug 2026 – Present)

- Research: Mechanical abuse and safety in Li-ion and Na-ion batteries; electro-chemo-mechanical modeling

University of Delaware – Ph.D. in Mechanical Engineering (Sep 2024 – Aug 2026)

- GPA: 4.0/4.0
- Research: Mechanical abuse and safety in Li-ion and Na-ion batteries; electro-chemo-mechanical modeling

Shanghai University – M.S. in Solid Mechanics (Sep 2021 – Jul 2024)

- GPA: 92.4/100
- Thesis: A pre-strain strategy of current collectors for suppressing electrode debonding in lithium-ion batteries

Shanghai Normal University – B.S. in Civil Engineering (Sep 2017 – Jul 2021)

- GPA: 2.93/4.0

Research Projects:

Electro-Chemo-Thermo-Mechanical Coupling Modeling (Abaqus Collaboration, 2025–Present)

- Developed Abaqus-based multiphysics models to simulate ball indentation–induced thermal runaway and aging of defective cells.

Mechanical Properties of Li-Ion Battery Materials (GM Collaboration, 2024.09–2025.12)

- Performed systematic tests on electrode and separator materials under varying temperature, state of charge (SOC), humidity, and strain rate conditions.

Publications & Presentations:

Citation: 125 (<https://scholar.google.com/citations?user=eAPEMPQAAAAJ&hl=en>)

Publications:

- [1] **Rui, B.**, Sun, S., Tan, X., Chak, C., Ma, L., Xu, J. Mechanical abuse and safety in sodium-ion batteries. *Journal of Materials Chemistry A*, 13, 12203–12215 (2025). (**Hot Paper**)
- [2] **RUI, B.**, HAHN, Y., BHOWMICK, A., MOUSAVI, F., KULATHU, S., OANCEA, V. and XU, J. Electrochemo-mechanical coupled behaviors in sodium-ion batteries upon stack pressure. *Journal of Materials Chemistry A* (2026)
- [3] **Rui, B.**, Lu, B., Song, Y., Zhang, J. A pre-strain strategy of current collectors for suppressing electrode debonding in lithium-ion batteries. *Applied Mathematics and Mechanics (English Edition)*, 44, 547–560 (2023).
- [4] SUN, S., **RUI, B.**, BAHUGUNA, S., ZHOU, J., SAYEED, F. and XU, J. Electrochemo-mechanical-thermal dynamics of internal short circuits in batteries. *Energy Storage Materials*, 105022 (2026) (**IF: 20.2**)
- [5] BHOWMICK, A., **RUI, B.**, SUN, S., TAN, X., WEI, B., HAHN, Y., KULATHU, S., OANCEA, V., LU, W. and XU, J. Electrochemo-mechanics unlocks hidden dynamics of lithium plating under stacking pressure. *Nano Energy*, **150**, 111751 (2026) (**IF: 17.1**)
- [6] Sun, S., **Rui, B.**, Tan, X., Bahuguna, S., Zhou, J., Xu, J. Effects of electrolyte, state of charge, and strain rate on the mechanical properties of lithium-ion battery electrodes and separators. *Journal of Materials Chemistry A*, 13, 20673–20687 (2025).
- [7] Xiong, Y., **Rui, B.**, Wang, S., Song, Y., Lu, B., Zhang, J. Safety-enhanced battery modules with actively switchable cooling and anti-impact functions. *International Journal of Mechanical Sciences*, 282, 109641 (2024).
- [8] Gao, H., Xiong, Y., **Rui, B.**, Bao, Y., Song, Y., Lu, B., Zhang, J. Chocolate pattern-inspired flexible thick electrodes: A facile mechanical imprinting method. *Chemical Engineering Journal*, 157266 (2024).
- [9] ZHU, K., SUN, S., **RUI, B.** and XU, J. Metal particle foreign matter as a hidden driver of battery aging: a defect-informed modeling study. *ACS Applied Materials & Interfaces* (2026)
- [10] Zhu, S., Lu, B., **Rui, B.**, Song, Y., Zhang, J. Revealing the quantitative connection between electrode-level cracks and capacity fading of silicon electrodes in lithium-ion batteries. *Electrochemistry*, 91, 127002 (2023).
- [11] SUN, S., BROWN, W., TAN, X., **RUI, B.**, XIONG, G., WANG, Y., ZENG, D. and XU, J. Aging matters: how degradation pathways govern thermal runaway in lithium-ion batteries. *Journal of Energy Chemistry*, 114, 10–21 (2026)
- [12] Tan, X., Zeng, N., Deng, Y., Sun, S., **Rui, B.**, Xu, J. Early-cycle prediction of battery aging onset across chemistries. *Journal of Energy Chemistry*, 105, 233–242 (2025).

Presentations (Conference talks/posters)

- **Rui, B.** Mechanical Abuse and Safety in Sodium-Ion Batteries. Battery Safety Workshop, Charlotte, NC, June 2025. (**Best Poster Award**)
- **Rui, B.** Mechanical Abuse and Safety in Sodium-Ion Batteries. Oral presentation, IMECE 2025.
- **Rui, B.** Electro-Chemo-Mechanical Coupling Unlocks New Insights into Lithium Plating and Aging. Oral presentation, IMECE 2025.
- **Rui, B.** Mechanism of Battery Lifespan Extension Upon Mechanical Pressure. Oral presentation, SAE 2026.

Professional Service & Leadership:

- **Co-Session Chair**, Electrochemical Energy Storage and Conversion Systems I (Session 09-01-01), IMECE 2025
- **Peer Reviewer: *5 in total***
 - 3 paper, Journal of Power Source
 - 1 paper, SAE Conference(2026)
 - 3 paper, IMECE Conference(2026)

Awards & Honors:

- Graduate Student Travel Award, University of Delaware (2025)
- Best Poster Award, Battery Safety Workshop Conference (2025)
- Outstanding Graduate Student of the Year, Shanghai University (2024)
- First Prize, National Scholarship (2023)
- Outstanding Presentation Award, Shanghai Mechanics Academic Conference (2022)
- Outstanding Student, Shanghai University (2023)
- Academic Scholarship, Shanghai University (2021, 2022, 2023)

Skills:

- **Experimental:** Dynamic mechanical testing, indentation, microscopy, materials characterization
- **Computational:** COMSOL Multiphysics, Abaqus, MATLAB, Origin, Python, High-performance computing (HPC)
- **Communication:** Academic writing, English/Chinese bilingual, conference presentations